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Register No.

Half-Yearly Examination - 2025

CHEMISTRY

Time : 3.00 Hrs. Marks : 70

PART - I

I. Note: (i) Answer all the questions. **15x1=15**

(ii) Choose the best answer and write the option code and the corresponding answer.

1. Considering Ellingham diagram, which of the following metals can be used to reduce alumina?
   a) Fe b) Cu c) Mg d) Zn
2. The element that does not show catenation among the following P-block elements.
   a) Carbon b) Silicon c) Lead d) Germanium
3. An element belongs to group 15 an 2nd period of the period table, its electronic configuration would be
   a) $1S^2 2S^2 2P^4$ b) $1S^2 2S^2 2P^3$ c) $1S^2 2S^2 2P^2$ d) $1S^2 2S^2 2P^5$
4. The magnetic moment of Mn^{4+} ion is a) 5.92 BM b) 2.80 BM c) 8.95 BM d) 3.90 BM
5. How many geometrical isomers are possible for $[Pt(Py)(NH_3)(Br)(Ce)]$? a) 3 b) 4 c) 0 d) 15
6. The yellow colour in NaCl crystal is due to
   a) excitation of electrons in F centres b) reflection of light from Cl ion on the surface
   c) reflection of light from Na^+ ion d) all of the above
7. If 75% of a first order reaction was completed in 60 minutes, 50% of the same reaction under the same conditions would be completed to
   a) 20 minutes b) 25 minutes c) 75 minutes d) 30 minutes.
8. Conjugated base for Bronsted acids H_2O and HF are
   a) OH an H_2FH^+ b) H_3O^+ and F^- repectively c) H_3O^+ and H_2FH^+ respectively d) OH^- and F^- respectively
9. How many Faradays of electricity are required for the following reactions to occur $MnO^{-4} \rightarrow Mn^{2+}$
   a) 3F b) 5F c) 1F d) 7F
10. Smoke is colloidal solution of a) liquid in gas b) solid in liquid c) gas in liquid d) gas in solid
11. Which one of the following is the strongest acid?
    a) 2-nitrophenol b) 4-chlorophenol c) 4-nitrophenol d) 3-nitrophenol
12. Assertion: Nylon-6.6 is a condensation polymer.
    Reason: Addition polymerization involves monomers directly adding to growing chain without formation of by-products.
    a) Assertion is true but reason is false b) Both assertion and reason are false
    c) Both assertion and reason are true but correct explanation d) both assertion and reason are true but not correct explanation
13. Insulin, a hormone chemically a) Fat b) Steroid c) Protein d) Carbohydrate
14. Which one of the following reduces Tollen's Reagent
    a) Formic acid b) Acetic acid c) Benzao phenone d) none of these
15. The product formed by the reaction an aldehyde with a primary amine
    a) Carboxylic acid b) Aromatic acid c) Schiff's base d) Ketone

PART - II

II. Note: Answer any six questions. Question No.24 is compulsory. **6x2=12**

16. What is calcination? Give examples.
17. How will you prepare bleaching powder?
18. Explain Frenkel defect with example.
19. Calculate the PH of 1.5×10^{-3} M solution of $Ba(OH)_2$

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20. Write a note on sacrificial production.
21. Write Kolbe's reaction.
22. Classify the following as linear branched or cross linked polymer.
    (a) Bakelite (b) Nylon-6.6 (c) LDPE (d) HDPE
23. Define Equivalent Conductance.
24. Identify A and B in the following sequence of reactions. $CH_3Br \xrightarrow{NaN_3} A \xrightarrow{LiAlH_4} B + N_2$

PART - III

III. Note: Answer any six questions. Question No.33 is compulsory. **6x3=18**

25. Explain Froth Floatation Method.
26. Give the uses of Helium.
27. Justify the position of lanthanides and actinides in the periodic table.
28. What is linkage isomerism? Explain with an example.
29. Aluminium crystallize in a cubic close packed structure. Its metallic radius Is 125 pm. Calculate the edge length of unit cell.
30. Differentiate physisorption and chemisorption.
31. Write Friedel Craft's acylation of anisole.
32. Write a note on zeolites.
33. Identify A and B in the following reactions.



PART - IV

IV. Note: Answer all the questions. **5x5=25**

34. a) Describe the role of the following in the process mentioned.
    1) Slica in the extraction of copper 2) Iodine in the refining of zirconium
    3) Give a reason to support that sulphuric acid is a dehydrating agent. **(OR)**
    b) Describe the structure of Diborane.
35. a) (1) What is lanthanoid contraction and what are the consequences of lanthanoid contraction.
    (2) Which is stronger reducing agent – Cr^{2+} or Fe^{2+} (OR)
    b) Write the oxidation state, coordination number, nature of ligand, magnetic property and electronic configuration of octahedral crystal field for the complex $K_4[Fe(CN)_6]$.
36. a)(i) Why ionic crystals are hard and brittle? (ii) Derive integrated rate law for a zero order reaction. $A \rightarrow$ product **(OR)**
    b) Derive an expression for Ostwald Dilution Law.
37. a) (i) State Faraday's laws of electrolysis. (ii) What are the factors affecting adsorption. **(OR)**
    b) Write a note on (i) Carbylamine reaction (ii) Gabriel Phthalimide Synthesis.
38. a) How will you convert:
    (i) Benezene → Benzaldehyde
    (ii) Ethene → Ethane 1,2,diol
    (iii) What are bio-degradable polymers? Give an example. **(OR)**
    b) (i) Short note on Peptide Bond.
    (ii) Different types of RNA which are found in cell.

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